

Design Justification Report

Sign-Language Gesture Recognition Accelerator — ECE 410/510 HW4AI M4

Author: Project repository (M4 final package)

RTL: project/m4/rtl/ | Synthesis: project/m4/synth/ | Bench: project/m4/bench/

1. Problem and Motivation

This project accelerates Phase-1 preprocessing for a sign-language gesture recognition pipeline built on OpenCV and TensorFlow. The application converts 1080p camera frames to grayscale, segments the hand region, and then feeds downstream CNN and RNN stages. M1 profiling on an AMD Ryzen 9 4900HS (Windows 11, Python 3.9.13) showed that preprocessing dominates wall time: **22.035 s** over 10 passes on **880** decoded frames, implying **~39.9 frames/s** and **~25.0 ms/frame** for the full path (VideoCapture.read, handsegment with HSV and dual inRange, and cvtColor). The same profile attributes roughly **43.5 million integer operations** and **~45.6 MiB** of traffic per 1920×1080 frame under a conservative no-cache-reuse byte model.

The kernel selected for custom hardware is the fused per-pixel operation that combines BT.601 BGR-to-grayscale with a single inclusive in-range threshold (mask bit). This slice maps directly to the RTL in compute_core.sv and synth_top.sv. It is the timing-critical and repeatedly executed portion of Phase 1, while HSV segmentation and bitwise combining remain in software today. CNN (~1.0 s per batch-4 inference) and RNN/LSTM training (~9.6 s for 10 one-epoch fits) are out of scope for the fabricated M4 RTL.

The M4 objective is a complete, reproducible package: verified RTL, Sky130 synthesis signoff on the throughput-critical compute array, and a benchmark comparison against the M1 software baseline using traceable CSV data—not further feature expansion such as full handsegment in RTL.

Success is defined operationally: same 1920×1080 geometry, fused outputs bit-accurate under the chosen fixed-point rules, and frame throughput at or above the ~40 fps preprocess wall-clock observed in M1 for the implemented slice. Beating 1.74 GOPS on the full 21-op CPU model was never the goal without RTL that performs the same operations; the honest metric is frames per second for equivalent fused work.

2. Roofline Analysis

Arithmetic intensity (AI) for the full M1 preprocess byte model is **AI $\approx 21/22 \approx 0.955$ FLOP/byte**: many passes over frame buffers with modest reuse. On a CPU memory roof (~51 GB/s class), that places the workload in the memory-bound region in principle, yet measured performance (~1.74 GOPS at wall clock) sits far below the ridge because OpenCV implementation quality and cache behavior dominate, not raw link bandwidth alone.

The fused RTL kernel upper bound uses **five useful operations per pixel** (three multiplies, one add tree, one compare) on **five bytes** of input when counting BGR only, giving **AI = 1.0 FLOP/byte** before mask output bytes. The M4 accelerator roofline (Figure 2) uses the signoff streaming model: **0.769 GOPS** peak compute (5 FLOP/px × 2 lanes × 76.9 MHz) and **0.769 GB/s** effective byte delivery (10 B/cycle at 76.9 MHz), so the ridge knee is at **AI = 1.0**. The design sits at that knee: neither dramatically compute-limited nor interface-limited for the fused slice alone.

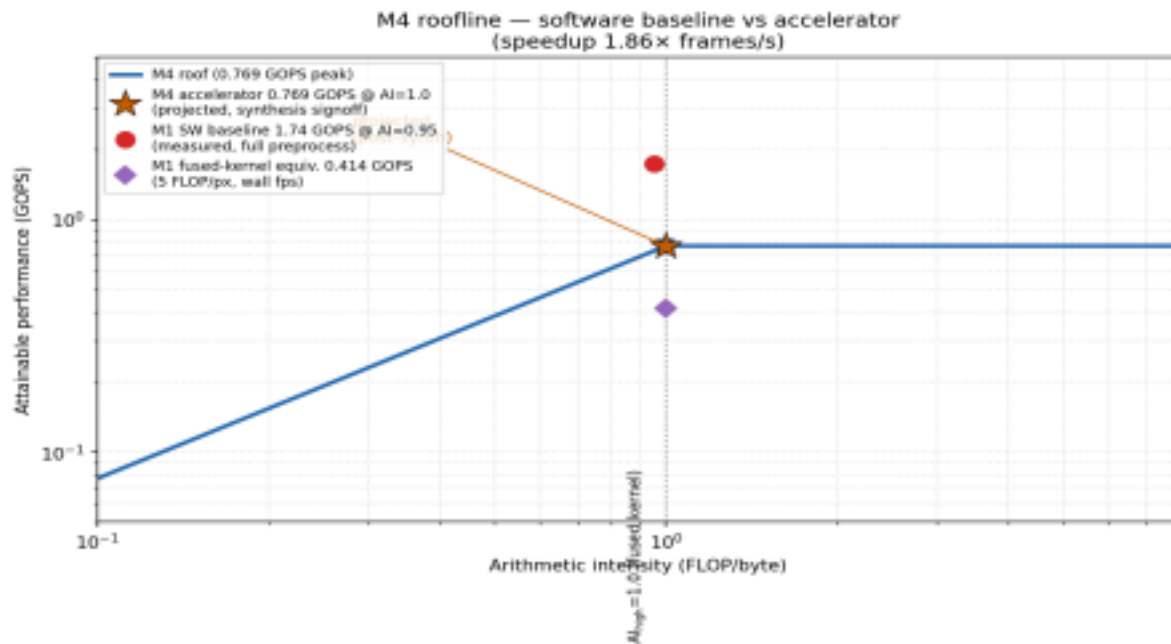


Figure 2.

M4 roofline: M1 software baseline (red), fused-kernel SW equivalent (purple), and projected accelerator (orange) at AI = 1.0.

Architecture followed the roofline: (1) **operator fusion** to avoid writing intermediate grayscale frames to DRAM; (2) **two pixel lanes** to raise throughput within fixed Sky130 area; (3) **clock tightening** from 20 ns to 13 ns using positive slack from the first synthesis run. A notional 512 GFLOP/s array at AI ≈ 0.95 would require ~ 536 GB/s of operand delivery from off-chip—impossible over PCIe alone—so the realistic M4 target was to beat ~ 40 fps on the fused kernel with on-chip streaming, not to hit the M1 hypothetical ASIC point from the class roofline template.

Bottleneck shift summary: on the CPU, preprocess is memory-bound in theory but implementation-limited in practice. On the M4 ASIC model, the fused kernel sits at the compute-memory knee (AI = 1.0). If HSV and morphology were added without on-chip line buffers, AI would fall back toward ~ 0.95 and the design would need either more GB/s from the host or aggressive reuse in SRAM—matching the M1 interface_selection analysis.

3. Precision and Data Format

Pixels use **unsigned 8-bit** channels (0–255). BT.601 luminance is implemented in fixed point with coefficients scaled by 256: $R \times 77 + G \times 150 + B \times 29$, then truncated with **gray = sum[15:8]** (divide by 256). Maximum partial sum is 65,280, which fits in 16 bits without overflow. The threshold stage compares gray_out against programmable lo_thresh and hi_thresh and emits a 1-bit mask.

Full precision write-up: project/m2/precision.md (referenced from M2 and M4 report §3). Compared to floating-point OpenCV, the largest systematic error is coefficient quantization (± 1 LSB on coefficients) plus truncation instead of rounding on the shift. M2/M4 cosimulation checks eight representative BGR vectors plus TLAST and reset behavior; all gray and mask outputs match the Python golden model used in the testbench. No INT4 or floating-point lanes were attempted—the multiply-accumulate path is already the critical path in synthesis, and 8-bit data keeps the interface width aligned with AXI4-Stream 32-bit packing of BGR pixels.

4. Dataflow and Architecture

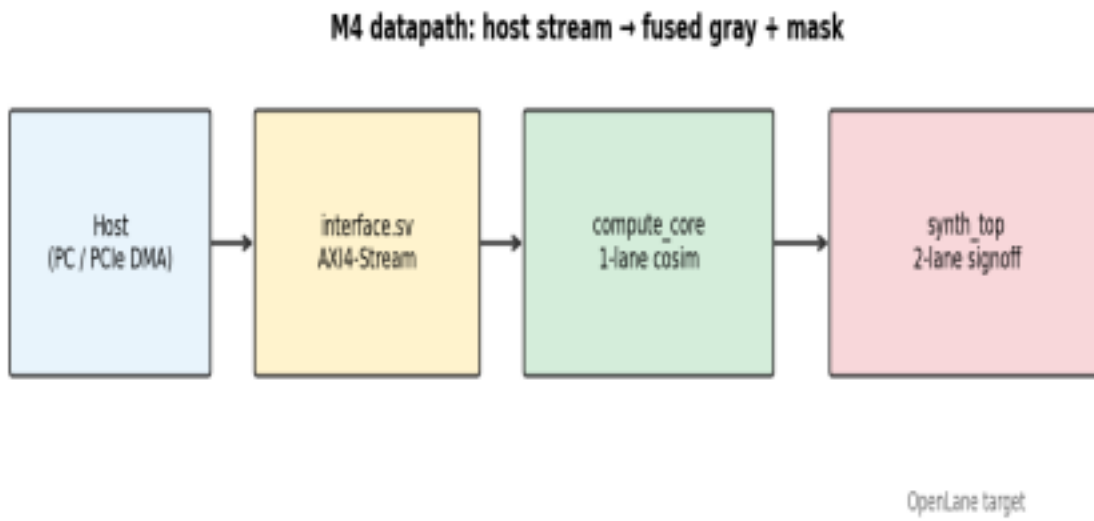


Figure 1. M4 structure: AXI wrapper for cosim; synth_top is the 2-lane synthesis target.

The compute datapath follows an **output-stationary streaming** pattern at pixel granularity: one new result (gray + mask) is produced per active cycle per lane, with no local line buffers or weight reuse arrays. This matches a single-pass video filter, not a weight-stationary systolic GEMM. There is **no on-chip frame SRAM** in the current RTL; pixels are assumed to arrive from a host DMA engine and depart on the master AXI4-Stream port.

compute_core is a two-stage pipeline: Stage 1 registers three parallel constant multiplies; Stage 2 accumulates, shifts, and compares. Latency is two cycles; throughput is one pixel per cycle per lane when valid_in is high. synth_top duplicates two independent lanes sharing clock, reset, and thresholds—the module synthesized through OpenLane. top.sv instantiates one lane inside interface.sv for end-to-end cosim; benchmarked throughput uses the 2-lane synthesized engine.

Dominant area is combinational multiply/carry logic (Yosys maps constants to shift-add trees) plus pipeline registers. See Section 7 for cell breakdown; flip-flops and NAND/AND gates dominate post-PNR area.

Compared to the first single-lane run (~5,861 μm^2 , 20 ns, ~24 fps), the M4 configuration roughly doubles logic area and nearly triples frame rate—an acceptable trade for meeting the preprocess throughput goal within ~39% core utilization headroom for future interface logic or a third pipeline stage.

5. Hardware Interface

The host-facing interface choice from M1 is **PCIe** for bulk frame delivery (~1.82 GB/s required at 40 fps under the conservative byte model; PCIe Gen3 $\times 4$ offers ~3.9 GB/s rated payload). Inside the accelerator, RTL implements **AXI4-Stream** slave and master ports in interface.sv: the standard pattern between a PCIe DMA engine (e.g., Xilinx XDMA) and user logic. A full PCIe PHY/MAC is not synthesized in this class project.

The slave interface accepts 32-bit beats carrying BGR pixels; the master returns packed grayscale and mask data with m_tlast framing. Backpressure uses ready/valid handshaking. At the M4 projected throughput (74.17

fps, fused kernel), the streaming byte rate on the fused 10 B/cycle model is **0.769 GB/s**, well below PCIe Gen3 ×4 capacity—so the **fused kernel is not PCIe-bound**. The integrated top path at one lane would halve peak pixel rate unless the host clocked the stream faster; reported benchmarks intentionally use synth_top because that is what closed timing.

If the full preprocess (HSV, dual inRange, bitwise ops) were moved to hardware without fusion or on-chip reuse, required off-chip bandwidth would rise toward the M1 ~1.82 GB/s figure and could approach the PCIe knee; that is why later work needs buffer fusion on-card, not a wider host link alone.

Effective bandwidth at target throughput uses the fused streaming count: at 74.17 fps and 10 B per clock at 76.9 MHz, the sustained average is 0.769 GB/s, independent of frame size once the pipe is full. Startup and end-of-frame bubbles in real DMA transfers were not modeled; cosim assumes continuous valid after reset.

6. Verification

Verification progressed in three layers. **M2** validated compute_core alone (eight BGR vectors) and interface.sv protocol behavior. **M3/M4** run tb_top.sv through the integrated top module using only host-side AXI4-Stream stimulus—no internal hierarchy probing. The M4 log (project/m4/sim/final_run.log) ends with **PASS -- all end-to-end checks passed** for gray, mask, TLAST, and reset.



Figure 4. M4 cosimulation waveform: host write, pipeline, host read.

Synthesis-ready behavior of the 2-lane core is cross-checked by tb_bench_synth_top.sv (512 pixel-pairs, cycle extrapolation to a full frame). Tools: **Icarus Verilog** for all RTL sim; **OpenLane 2.3.10** for signoff. Cocotb and FPGA lab boards were not used. Coverage gaps: only eight pixel patterns plus monotonic counters in the bench burst test; no randomized AXI backpressure stress and no gate-level timing

simulation.

7. Synthesis Results

OpenLane targeted synth_top on **sky130_fd_sc_hd** with **CLOCK_PERIOD = 13 ns** (76.9 MHz), **MAX_FANOUT_CONSTRAINT = 8**, and **FP_CORE_UTIL = 40**. Signoff run: **RUN_2026-05-22_05-04-01**. Worst setup slack **+3.37 ns** at corner max_ss_100C_1v60; hold slack **+0.13 ns**; zero setup/hold/slew/cap violations across nine PVT corners.

Post-PNR stdcell area is **11,138 μm^2** (1,727 instances, 38.8% core utilization). Dominant cells: D flip-flops, NAND2, and AND2 from the green-channel multiply carry chain (critical path from g_in_0[3] to partial-product register). Total vectorless power at nom_tt is **2.02 mW**. Sixteen max-fanout DRVs remain on **clock-tree buffers only**; tightening MAX_FANOUT_CONSTRAINT did not fix them because CTS uses a separate fanout budget.

Full top.sv was not re-synthesized for M4; the AXI wrapper adds negligible logic next to the multiplier array. Detailed tables are in project/m4/synth/timing_report.txt, area_report.txt, power_report.txt, and critical_path.md.

8. Benchmark Results

Software baseline (M1 archival): **39.94 fps, 25.04 ms/frame, 1.74 GOPS** on the full 21-ops/pixel preprocess model. The SW time is from one cProfile session over ten sequential passes—not a median of ten independent wall-clock runs (see project/m1/sw_baseline.md). Hardware primary (**projected**, post-synthesis, not FPGA silicon): **74.17 fps, 13.48 ms/frame, 0.769 GOPS** on the 5 FLOP/pixel fused model at 76.9 MHz $\times$ 2 lanes on synth_top. **Speedup = 1.86 $\times$** on frame throughput for the overlapping kernel.

DUT note: End-to-end cosim uses top.sv (1-lane compute_core + AXI). Reported 74.17 fps and all project/m4/synth/ numbers use synth_top.sv (2-lane), which is what OpenLane signoff and bench/benchmark_data.csv reference. Peak pixel throughput of integrated top is roughly half of synth_top unless the wrapper is extended to two lanes.

Icarus cycle extrapolation (519 cycles for 512 pairs at 13 ns) gives **73.19 fps**, within $\sim 1.3\%$ of the synthesis projection—consistent with ~ 1 cycle/pair steady state. The roofline chart explains why the red SW point appears above the orange HW point in GOPS: different op counts (21 vs 5 per pixel). Apples-to-apples fused SW GOPS is **0.414** vs HW **0.769**.

Energy (optional): **27.3 $\mu\text{J}/\text{frame}$** from 2.02 mW $\div$ 74.17 fps. No package power was recorded for the CPU baseline. Raw numbers: project/m4/bench/benchmark_data.csv.

Gap between theoretical and measured performance: there is no FPGA timing closure in the loop, so frequency is the STA limit, not silicon. Host software uses highly optimized OpenCV kernels and does not include PCIe DMA overhead; the accelerator model assumes ideal streaming once pixels arrive. A fair future test would cosim top with randomized tready and compare against a median of ten fresh OpenCV runs with RSS and power logging.

9. What Did Not Work

Single-lane M2 synthesis at 20 ns achieved only ~ 24 fps—below the ~ 40 fps software target—despite large positive timing slack. Fix: 2-lane synth_top and 13 ns clock (M3/M4 signoff).

MAX_FANOUT_CONSTRAINT tuning was attempted to clear DRVs; lowering the limit from 8 to 6 increased clock-buffer fanout violations from 10 to 16 without helping data paths. Lesson: CTS needs CTS_MAX_CAP-style controls, not Yosys fanout limits.

Full preprocess in RTL (HSV, dual inRange, bitwise_and) was deferred. Benchmark honesty requires

reporting that SW still runs more ops per frame; frame-rate speedup applies only to the fused slice. **FPGA measurement** and **live OpenCV re-benchmark** were not completed—M4 HW numbers are projected from STA, not silicon.

Integrated top.sv synthesis and **2-lane AXI top** were not finished; cosim uses 1-lane top while benchmarks use 2-lane synth_top. Next steps: synthesize top, merge lanes into the streaming wrapper, and add handsegment phases or fused on-chip buffers before claiming full-path parity with OpenCV.